Parsa Rangriz

Contact Information

Department of Mathematics,
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Education

University of California, San Diego 2025 - Present
PhD in Mathematics

University of Waterloo 2023 - 2025
MMath in Statistics
Advisor: Aukosh Jagannath
Thesis: High-dimensional scaling limits of online SGD in single-index models

Sharif University of Technology 2018 - 2023
BSc in Physics
Extra: Minor in Mathematics

Publications

Parsa Rangriz, *Limit Theorems for Stochastic Gradient Descent in High-Dimensional Single-Layer Networks*, Electronic Communications in Probability, 31 (2026).

Parsa Rangriz, *High-Dimensional Scaling Limits of Online Stochastic Gradient Descent in Single-Index Models*, MMath Thesis, University of Waterloo (2025).

Employment

Graduate Research Assistant. University of Waterloo 2023 - 2025
Supervisor: Aukosh Jagannath

Worked on high-dimensional probability, scaling limits, random matrices, and mathematical foundations of machine learning.

Summer Research Intern. EPFL 2022
Supervisor: Lenka Zdeborova

Worked on statistical physics methods for machine learning and combinatorial optimization, e.g., mean-field theory, replica method, and message-passing algorithms.

Awards & Honors

First-Year Departmental Fellowship. UC San Diego 2025 - 2026

Graduate Research Studentship. University of Waterloo 2023 - 2025

International Master's Award of Excellence. University of Waterloo 2023

Master of Math Entrance Scholarship. University of Waterloo 2023

Summer@EPFL Fellowship. EPFL 2022

Silver Medal in the 30th **Iranian National Physics Olympiad** 2017

TA Experience

UC San Diego (2025-Present)

PHYS 1CL: Waves, Optics & Modern Physics Lab (Summer 2026); MATH 193A: Actuarial Mathematics (Summer 2026); MATH 180A: Introduction to Probability (Spring 2026); MATH 10B: Calculus II (Winter 2026); MATH 11: Calculus-Based Probability and Statistics (Fall 2025)

University of Waterloo (2023-2025)

STAT 333: Stochastic Processes I (Winter 2025, Winter 2024); STAT 433: Stochastic Processes II (Fall 2024); STAT 330: Mathematical Statistics (Fall 2023, Spring 2024); STAT 231: Probability (Spring 2024); STAT 230: Statistics (Spring 2024); STAT 202: Introductory Statistics for Scientists (Fall 2023)